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Synthesis of allenynes *via* Pd-catalyzed coupling of 1,4-diyn-3-yl carbonates with boronic acids†

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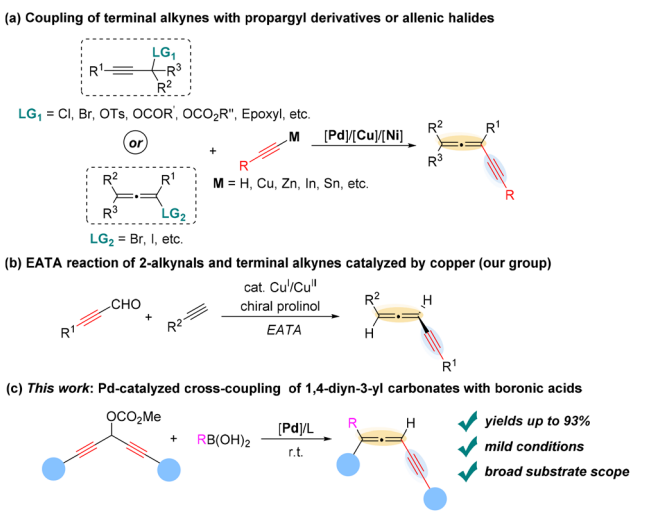
Conjugated allenynes widely exist in natural products and are commonly employed in the development of functional materials. However, synthetic methods are very limited. In this contribution, a palladium-catalyzed cross-coupling reaction of 1,4-diyn-3-yl carbonates with a series of commercially available boronic acids using S-phos or Gorlos-phos as the ligand has been developed. Diversely substituted allenynes incorporating synthetically useful functional groups are obtained in good yields under this set of mild conditions. Gram-scale reactions and synthetic transformations have been conducted to show their potential utility.

Introduction

Allenynes are typical units existing in some biologically active natural products,[1] such as marasin[2] and mycomycin,[3] with high antibiotic activity. In addition, conjugated allenynes are not only key structures in various functional materials[4] but also unique reactive intermediates in organic synthesis.[5] However, approaches towards allenynes are still limited. Palladium-catalyzed cross-coupling reactions of terminal alkynes or alkynyl organometallic reagents with propargylic derivatives or allenic halides have been developed (Scheme 1a).[6] Recently, our group realized the EATA (enantioselective allenation of terminal alkyne) reaction to afford various conjugated chiral allenynes with excellent enantioselectivity and regioselectivity: a catalytic reaction of copper halides with mixed oxidation states in the presence of commercially available (*S*)- or (*R*)-α,α-diphenylprolinol (Scheme 1b).[7] Here, we wish to report a new general method for the synthesis of conjugated allenynes: Pd-catalyzed cross-coupling reaction of 1,4-diyn-3-yl carbonates with commercially available boronic acids using S-phos or Gorlos-phos as the ligand under mild conditions, which may tolerate various synthetically useful functional groups (Scheme 1c).

Results and discussion

Initially, the ligand effect of the coupling reaction was investigated (Table 1). We chose 1,4-diyn-3-yl carbonate **1a** and 1.1 equiv. of (3-methoxyphenyl)boronic acid **2a** as the model substrates using $Pd_2(dba)_3{\cdot}CHCl_3$ as the catalyst in 1,4-dioxane at room temperature in the presence of H_2O (2 equiv.) for 12 hours. Among the typical monophosphine ligands, PCy_3 afforded the desired allenyne **3aa** in 59% NMR yield with 19% recovery of **1a** (entry 1). When PPh_3 was employed, no expected product **3aa** was formed with 91% recovery of **1a** (entry 2). When tri(tolyl)phosphines were employed, tri(*o*-methylphenyl) phosphine could furnish the corresponding product **3aa** in



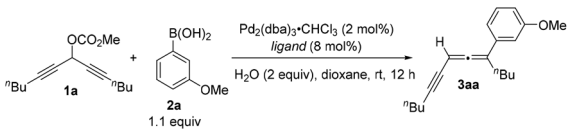
Scheme 1 Approaches towards conjugated allenynes catalyzed by transition metals.

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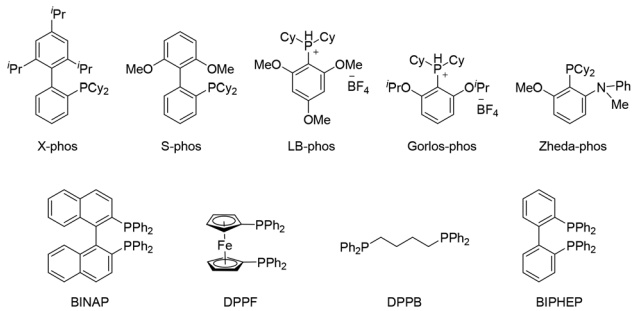
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Table 1 Optimization of ligands[a]



Entry	Ligand	NMR yield[b] of **3aa**/%	Recovery[b] of **1a**/%
1	PCy_3	59	19
2	PPh_3	0	91
3	$(2\text{-}MeC_6H_4)_3P$	83	0
4	$(4\text{-}MeC_6H_4)_3P$	19	66
5	$(2\text{-}OHCC_6H_4)Ph_2P$	0	90
6	X-phos	74	14
7	**S-phos**	**87**	**0**
8[c]	LB-phos	0	92
9[c]	Gorlos-phos	74	18
10	Zheda-phos	0	95
11[d]	BINAP	0	77
12[d]	DPPF	0	90
13[d]	DPPB	0	88
14[d]	BIPHEP	0	90

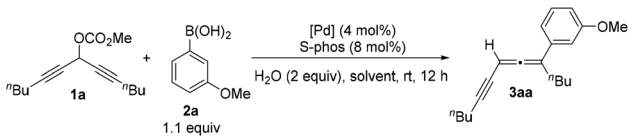
[a] Reaction conditions: **1a** (0.1 mmol), **2a** (0.11 mmol), $Pd_2(dba)_3{\cdot}CHCl_3$ (2 mol%), ligand (8 mol%), and H_2O (2 equiv.) in dioxane (0.5 mL) at room temperature for 12 h unless otherwise noted. [b] Determined by ^1H NMR analysis using dibromomethane as the internal standard. [c] 8 mol% K_2CO_3 was added. [d] 4 mol% ligand was used.



83% yield (entry 3); tri(*p*-methylphenyl)phosphine did not perform well (entry 4). Relatively electron-deficient monophosphine, *o*-(diphenylphosphino)benzaldehyde, led to 90% recovery of **1a** (entry 5). Then we found that X-phos could also deliver **3aa** in 74% yield with 14% recovery (entry 6); a high yield of 87% was achieved with no recovery when S-phos was used (entry 7). LB-phos,[8] Gorlos-phos,[9] and Zheda-phos[10] developed by our group were also tested (entries 8–10) and Gorlos-phos was found to be compatible with this cross-coupling reaction in a yield of 74%. Common bisphosphine ligands BINAP, DPPF, DPPB, and BIPHEP failed to give the allenyne product **3aa**.

We next screened various palladium catalysts and solvent with S-phos as the ligand (Table 2). The reaction led to 88% recovery of **1a** when $Pd_2(dba)_3{\cdot}CHCl_3$ was replaced with $Pd(PPh_3)_4$ (entry 2). The yield of the product was improved to 89% yield when $Pd(dba)_2$ was employed (entry 3), but $Pd(dmdba)_2$ afforded product **3aa** with lower efficiency (entry 4). Among the Pd(II) catalyst, only $Pd(OAc)_2$ could afford allenyne product **3aa** in a poor yield (entries 5–7).

Table 2 Optimization of the Pd catalyst and solvent[a]



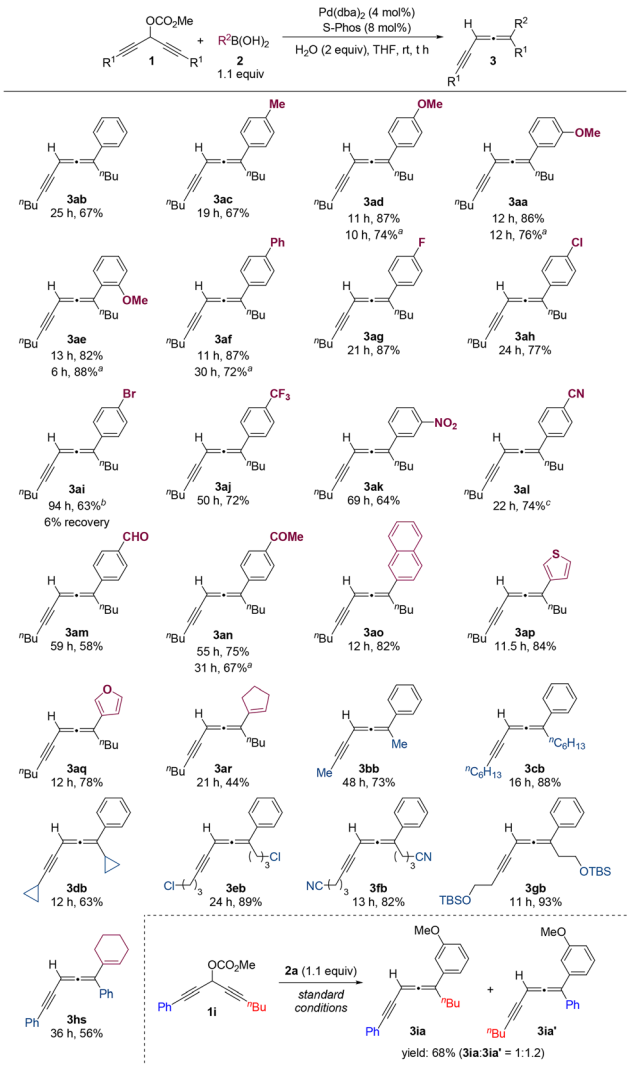
Entry	[Pd]	Solvent	NMR yield[b] of **3aa**/%	Recovery[b] of **1a**/%
1	$Pd_2(dba)_3{\cdot}CHCl_3$	Dioxane	87	0
2	$Pd(PPh_3)_4$	Dioxane	0	88
3	$Pd(dba)_2$	Dioxane	89	0
4	$Pd(dmdba)_2$	Dioxane	76	11
5	$PdCl_2$	Dioxane	0	85
6	$Pd(OAc)_2$	Dioxane	38	46
7	$[Pd(allyl)Cl]_2$	Dioxane	0	90
8	$Pd(dba)_2$	DCM	9	82
9	**$Pd(dba)_2$**	**THF**	**97**	**0**
10	$Pd(dba)_2$	Et_2O	88	0
11	$Pd(dba)_2$	EA	96	0
12	**$Pd(dba)_2$**	**Toluene**	**97**	**0**
13[c]	$Pd(dba)_2$	THF	65	35
14[d]	$Pd(dba)_2$	THF	36	59

[a] Reaction conditions: **1a** (0.1 mmol), **2a** (0.11 mmol), [Pd] (4 mol%), S-phos (8 mol%), and H_2O (2 equiv.) in solvent (0.5 mL) at room temperature for 12 h unless otherwise noted. [b] Determined by ^1H NMR analysis using dibromomethane as the internal standard. [c] 2 mol% $Pd(dba)_2$ was used. [d] Without H_2O.

With $Pd(dba)_2$ as the catalyst, the solvent effect was next investigated. The reaction was inhibited when DCM was used as solvent (entry 8). To our delight, THF and toluene could afford allenyne **3aa** in almost quantitative yields (entries 9 and 12). The yield decreased to 65% or 36% when reducing the loading of the catalyst or removing H_2O. Therefore, the optimal reaction conditions were defined as 4 mol% $Pd(dba)_2$, 8 mol% S-phos, and 2 equiv. of H_2O in THF at room temperature.

Having established the optimal conditions, the generality of this Pd-catalyzed cross-coupling reaction was examined (Scheme 2). First of all, a series of commercially available arylboronic acids were reacted with 1,4-diyn-3-yl carbonate **1a**. Besides phenylboronic acid, the aryl group may also be substituted with electron-donating groups, such as methyl, methoxyl, or phenyl at the *p*-position, smoothly forming allenynes **3ab–3ad** and **3af** in moderate to high yields. The steric effect is not obvious since *o*-, *m*- and *p*-methoxy-substituted phenyl boronic acids all afforded **3ae**, **3aa**, and **3ad** in >80% yields. Synthetically useful *p*-fluoro, *p*-chloro and *p*-bromo-substituted phenylboronic acids could all afford the corresponding products **3ag–3ai** in decent yields of 63–87%. Electron-deficient *p*-trifluoromethyl (**3aj**), *m*-nitro (**3ak**), *p*-cyano (**3al**), *p*-formyl (**3am**), and *p*-acetyl (**3an**) may all be tolerated. In addition, 2-naphthyl, 3-thienyl, and 3-furanyl-substituted boronic acids all worked to afford allenynes **3ao–3aq** in decent yields. Furthermore, cyclopentenyl boronic acid may also be applied affording product **3ar** in 44% yield. With phenylboronic acid, R^1 may be different carbon chains (C_1, C_6 and cyclopropyl) and halo-, cyano-, silylether-substituted alkyl groups forming

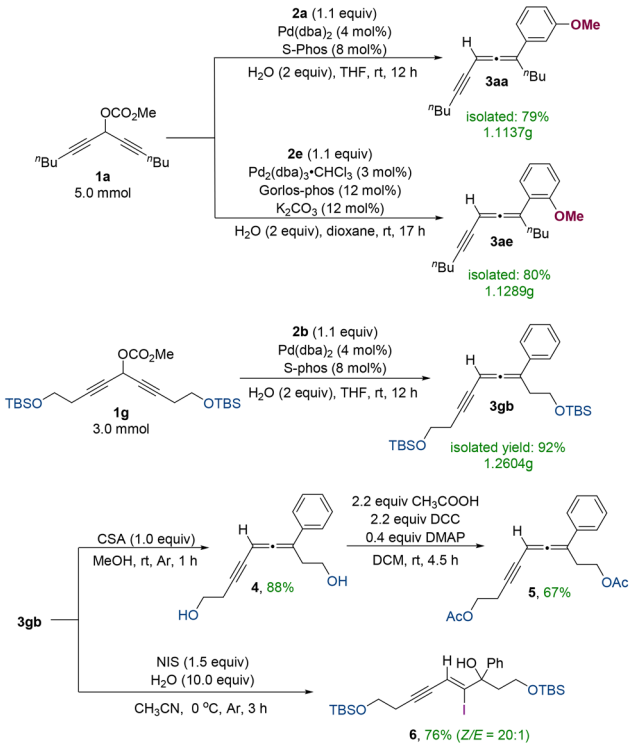
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Scheme 2 The substrate scope. Reaction conditions: **1** (1.0 mmol), **2** (1.1 mmol), $Pd(dba)_2$ (4 mol%), S-phos (8 mol%), and H_2O (2 equiv.) in THF (5.0 mL) at room temperature unless otherwise noted. [a] **1a** (1.0 mmol), **2** (1.1 mmol), $Pd_2(dba)_3{\cdot}CHCl_3$ (3 mol%), Gorlos-phos (12 mol%), K_2CO_3 (12 mol%), and H_2O (2 equiv.) in dioxane (5.0 mL) at room temperature. [b] 8 mol% $Pd(dba)_2$ and 16 mol% S-phos were used. [c] 40 °C.

allenynes **3bb**–**3gb** in 63–93% yields. R^1 may also be phenyl, affording **3hs** in 56% yield. Gorlos-phos was also tested in 1 mmol reactions, affording the desired products **3aa**, **3ad**–**3af**, and **3an** in 67–88% yields. To further expand the scope of allenyne, non-symmetric 1,4-diyn-3-yl carbonate **1i** was employed under the standard conditions. The reaction proceeded smoothly, however, the regioselectivity was not satisfactory with the formation of **3ia** and **3ia′** in a 1 : 1.2 ratio.

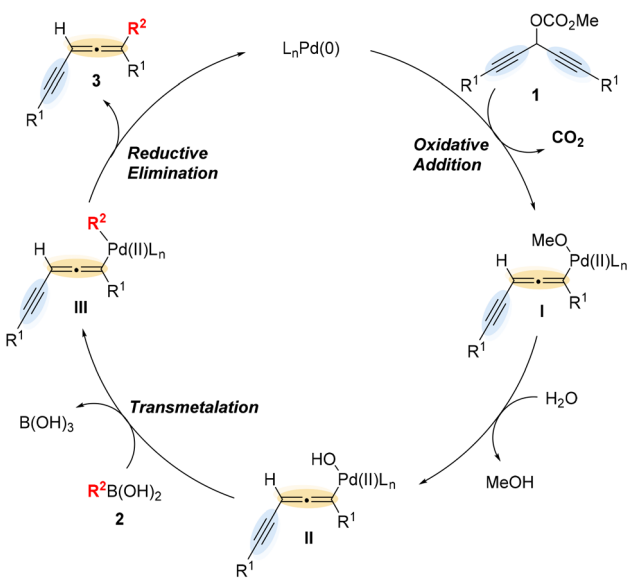
Three different reactions on a gram-scale were conducted under the standard conditions to afford the target products **3aa**, **3ae**, and **3gb** in 79–92% yields (Scheme 3). The synthetic potential of compound **3gb** has been demonstrated: the protecting group TBS could be easily removed with camphorsulfonic acid affording diol **4** in 88% yield[11] and the subsequent



Scheme 3 Gram-scale reactions and transformations.

esterification formed ester **5**; stereoselective iodohydroxylation of **3gb** with NIS and H_2O could deliver the useful intermediate 2-iodoallyl alcohol **6** in 76% yield (Z/E = 20 : 1).[12]

A catalytic cycle has been proposed as shown in Scheme 4. $L_nPd(0)$ coordinated with the S-phos or Gorlos-phos ligand would undergo S_N2'-type oxidative addition with 1,4-diyn-3-yl carbonate **1** to generate the allenylpalladium intermediate **I**



Scheme 4 The proposed mechanism.

after releasing a molecule of CO_2. Then the methoxy anion would be protonated with water to form allenylpalladium hydroxide intermediate **II**,[13] which would more efficiently undergo transmetalation with boronic acid **2** to generate intermediate **III**. Finally, the desired product allenyne **3** would be afforded after reductive elimination with the regeneration of the catalytically active species $L_nPd(0)$.

Conclusions

In summary, we have developed a Pd-catalyzed cross-coupling reaction of 1,4-diyn-3-yl carbonates with a series of commercially available boronic acids with S-phos or Gorlos-phos as the ligand. Under this set of mild reaction conditions, a variety of allenynes with synthetically useful functional groups are obtained in good yields. Three gram-scale reactions and the synthetic potential of the products have been demonstrated. Further studies on this topic are currently underway in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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References

1 A. Hoffmann-Röder and N. Krause, Synthesis and properties of allenic natural products and pharmaceuticals, *Angew. Chem., Int. Ed.*, 2004, **43**, 1196–1216.

2 (*a*) G. Bendz, Marasin, an antibiotic polyacetylene, isolated from the culture medium of marasmius ramealis, *Ark. Kemi*, 1959, **14**, 305–321; (*b*) R. C. Cambie, A. Hirschberg, E. R. H. Jones and G. Lowe, Chemistry of the higher fungi. Part XVI. Polyacetylenic metabolites from aleurodiscus roseus, *J. Chem. Soc.*, 1963, 4120–4130.

3 W. D. Celmer and I. A. Solomons, The structure of the antibiotic mycomycin, *J. Am. Chem. Soc.*, 1952, **74**, 1870–1871.

4 P. Rivera-Fuentes and F. Diederich, Allenes in molecular materials, *Angew. Chem., Int. Ed.*, 2012, **51**, 2818–2828.

5 For selected examples of allenynes in organic synthesis, see: (*a*) A. P. Khrimyan, A. V. Karapetyan and Sh. O. Badanyan, Regioselective addition of hydrazines to allenylacetylenes – convenient route to 3(5)-methyl-5(3)-substituted pyrazoles, *Chem. Heterocycl. Compd.*, 1984, **20**, 189–196; (*b*) S. Kim and P. H. Lee, Cyclization of allenyne-1,6-diols catalyzed by gold and silver salts: an efficient selective synthesis of dihydrofuran and furan derivatives, *Adv. Synth. Catal.*, 2008, **350**, 547–551; (*c*) C. M. Reisinger, P. Rivera-Fuentes, S. Lampart, W. B. Schweizer and F. Diederich, Cascade pericyclic reactions of alleno-acetylenes: facile access to highly substituted cyclobutene, dendralene, pentalene, and indene skeletons, *Chem. – Eur. J.*, 2011, **17**, 12906–12911; (*d*) T. Wang, R. R. Naredla, S. K. Thompson and T. R. Hoye, The pentadehydro-Diels-Alder reaction, *Nature*, 2016, **532**, 484–488; (*e*) X. Li, H. Yu and Y. Huang, Amine-triggered 6π-electrocyclization-aromatization cascade of ynedienamines, *Adv. Synth. Catal.*, 2017, **359**, 1379–1387; (*f*) A. Le, S. Gupta, M. Xu, Y. Xia and D. Lee, Development of an allenyne-alkyne [4+2] cycloaddition and its application to total synthesis of Selaginpulvilin A, *Chem. – Eur. J.*, 2022, **28**, e202202015; (*g*) Q. Xu and T. R. Hoye, Electronic character of α,3-dehydrotoluene intermediates generated from isolable allenyne-containing substrates, *Angew. Chem., Int. Ed.*, 2022, **61**, e202207510.

6 For selected examples, from allenic halides to conjugated allenynes, see: (*a*) K. Ruitenberg, H. Kleijn, C. J. Elsevier, J. Meijer and P. Vermeer, Palladium(0)-promoted synthesis of functionally substituted allenes by means of organozinc compounds, *Tetrahedron Lett.*, 1981, **22**, 1451–1452; (*b*) T. Jeffery-Luong and G. Linstrumelle, Palladium-catalyzed synthesis of allenynes, *Synthesis*, 1983, 32–34; (*c*) A. M. Caporusso, F. D. Settimo and L. Lardicci, Metal catalysis in organic reactions. 17. A nickel-promoted route to substituted allenes by reaction of 1-bromo-1,2-dienes with alkyl metals, *Tetrahedron Lett.*, 1985, **26**, 5101–5104; (*d*) R. W. Saalfrank, M. Haubner, C. Deutscher and W. Bauer, Yne-allenes from 1-bromoallenes and trimethylstannylacetylenes via palladium-mediated C-C coupling reactions, *Eur. J. Org. Chem.*, 1999, 2367–2372. From propargylic derivatives to conjugated allenynes, see: (*e*) T. Mandai, T. Nakata, H. Murayama, H. Yamaoki, M. Ogawa, M. Kawada and J. Tsuji, Palladium-catalyzed reactions of 2-alkynyl carbonates with terminal acetylenes: a new synthetic method for 1,2-dien-4-ynes, *Tetrahedron Lett.*, 1990, **31**, 7179–7180; (*f*) S. Gueugnot and G. Linstrumelle, An efficient palladium-catalysed reaction of propargyl halides, tosylates and acetates with terminal alkynes, *Tetrahedron Lett.*, 1993, **34**, 3853–3856; (*g*) C. Darcel, C. Bruneau and P. H. Dixneuf, Palladium(0), copper(I) catalysed synthesis of conjugated alkynyl α-allenols from alkynyl cyclic carbonates and terminal alkynes, *J. Chem. Soc., Chem. Commun.*, 1994, 1845–1846; (*h*) S. Condon-Gueugnot and G. Linstrumelle, Palladium-catalyzed regioselective coupling of propargylic substrates with terminal alkynes. Application to the synthesis of 1,2-dien-4-ynes, *Tetrahedron*, 2000, **56**, 1851–1857; (*i*) P. Bertus, F. Fécourt, C. Bauderb and P. Pale, Evidence for the *in situ* formation of copper acetylides during Pd/Cu catalyzed synthesis of enynes: a new synthesis of allenynols, *New J. Chem.*, 2004, **28**, 12–14; (*j*) R. Riveiros, D. Rodríguez, J. P. Sestelo and L. A. Sarandeses, Palladium-catalyzed

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cross-coupling reaction of triorganoindium reagents with propargylic esters, *Org. Lett.*, 2006, **8**, 1403–1406; (*k*) M. Yoshida, M. Hayashi and K. Shishido, Palladium-catalyzed diastereoselective coupling of propargylic oxiranes with terminal alkynes, *Org. Lett.*, 2007, **9**, 1643–1646; (*l*) D. Girard, S. Broussous, O. Provot, J.-D. Brion and M. Alami, Palladium mediated direct coupling of silylated arylalkynes with propargylic chlorides: an efficient access to functionalized conjugated allenynes, *Tetrahedron Lett.*, 2007, **48**, 6022–6026; (*m*) M. K. Smith and J. A. Tunge, Palladium-catalyzed synthesis of conjugated allenynes via decarboxylative coupling, *Org. Lett.*, 2017, **19**, 5497–5500. Other examples: (*n*) D. Taç and L. Artok, Palladium-catalyzed coupling of 2-en-4-yne carbonates with terminal alkynes, *Tetrahedron Lett.*, 2018, **59**, 895–898; (*o*) X.-Y. Dong, T.-Y. Zhan, S.-P. Jiang, X.-D. Liu, L. Ye, Z.-L. Li, Q.-S. Gu and X.-Y. Liu, Copper-catalyzed asymmetric coupling of allenyl radicals with terminal alkynes to access tetrasubstituted allenes, *Angew. Chem., Int. Ed.*, 2021, **60**, 2160–2164; (*p*) W. Liu, C. Liu, M. Wang and W. Kong, Modular synthesis of multifunctionalized CF_3-allenes through selective activation of saturated hydrocarbons, *ACS Catal.*, 2022, **12**, 10207–10221; (*q*) H. Huang, H. Zhang, Q. Wang, Y. Sun, L. Su, W. Xu, Y. Ma, S. Kong, G. Zhang and R. Guo, Copper-catalyzed asymmetric 1,4-aryl/alkynylation of 1,3-enynes to access axially chiral tetrasubstituted allenes, *ChemCatChem*, 2023, **15**, e202300697.

7 G. Wu, Y. Yao, G. Li, X. Zhang, H. Qian and S. Ma, Enantioselective allenation of terminal alkynes catalyzed by copper halides of mixed oxidation states and its application to the total synthesis of Scorodonin, *Angew. Chem., Int. Ed.*, 2021, **60**, 2–10.

8 B. Lü, C. Fu and S. Ma, Application of a readily available and air stable monophosphine HBF_4 salt for the Suzuki coupling reaction of aryl or 1-alkenyl chlorides, *Tetrahedron Lett.*, 2010, **51**, 1284–1286.

9 B. Lü, P. Li, C. Fu, L. Xue, Z. Lin and S. Ma, 2,6-Diisopropoxyphenyl(dicyclohexyl)phosphine: A new ligand for palladium-catalyzed amination reactions of aryl chlorides with potassium hydroxide as the base, *Adv. Synth. Catal.*, 2011, **353**, 100–112.

10 P. Li, B. Lü, C. Fu and S. Ma, Zheda-Phos for general α-monoarylation of acetone with aryl chlorides, *Adv. Synth. Catal.*, 2013, **355**, 1255–1259.

11 S. Umemiya and M. Terada, Catalytic enantioselective allylation of acetylenic aldehydes by chiral phosphoric acid/transition metal cooperative catalysis: formal synthesis of Fostriecin, *Org. Lett.*, 2021, **23**, 3767–3771.

12 W. Kong, B. Guo, C. Fu and S. Ma, An efficient approach to 2-bromoalken-3-ols by regioselective bromohydroxylation reaction of simple allenes with NBS, *Eur. J. Org. Chem.*, 2011, 2278–2285.

13 For selected examples, see: (*a*) J. Xiao, H. Luo, S. Huang, H. Qian and S. Ma, Tri(*o*-tolyl)phosphine for highly efficient Suzuki coupling of propargylic carbonates with boronic acids, *Chem. Commun.*, 2018, **54**, 10451–10454; (*b*) C. Amatore, A. Jutand and G. Le Duc, Kinetic data for the transmetalation/reductive elimination in palladium-catalyzed Suzuki-Miyaura reactions: unexpected triple role of hydroxide ions used as base, *Chem. – Eur. J.*, 2011, **17**, 2492–2503; (*c*) B. P. Carrow and J. F. Hartwig, Distinguishing between pathways for transmetalation in Suzuki-Miyaura reactions, *J. Am. Chem. Soc.*, 2011, **133**, 2116–2119.

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